

S10 — adversarial review, with responses

Updated 2026-08-03. Twelve findings from the adversarial methodology review, each with the attack stated plainly, our response, and what has been done. Terms of art are marked with a dagger† on first use and defined in §5.

Three findings were re-derived independently from the raw data before being accepted, because a review reporting 48 of 49 findings surviving its own verification is itself a thing to be suspicious of. All three reproduced.

Since the first version of this document, two findings have been obviated rather than fixed — the analyses they attacked have been removed from the study. That is the cleaner outcome where it is available, and §2 says why.

1. Status

#	Finding	Severity	Status
F1	The word-level mechanism test is a rarity index, not an AI-preference score	Fatal	OBVIATED — analysis and its corpus deleted
F3	Kobak's content words were never run, and v1.0 fails them	Major	OBVIATED — control cut as orthogonal
F5a	Sensitivity measured on one 7B generator	Major	OBVIATED — replaced by published evaluations
F2	Placebo pool† matched on frequency but not dispersion†	Major	Applied (protocol v1.1)
F4	The NB prevalence figure is a five-month window; §5 names the wrong stratifier†	Major	Applied
F6	The frequency-weighted secondary does not replicate; the sweep claim is misquoted	Major	Applied — full table now printed
F7	The formality control is not actually frequency-matched	Major	Applied — buckets now fill
F8	One ratio pooled over 2024–26 is not a stable estimand†	Major	Applied — and it exposed instrument decay
F9	Zero-count smoothing† is asymmetric between instrument and placebo	Minor	Applied — now load-bearing (see below)
F11	Instrument citation wrong (fourth author is Lause, not Berens)	Minor	Applied
F12	No-interference† assumption never stated	Minor	Applied
F5b	Sensitivity on <i>edited</i> text is unmeasured by anyone	Major	Stands — unfixable; stated as a floor
F10	Estimand departs from Kobak's, forfeiting the prevalence reading	Minor	OUTSTANDING

Net effect on the headline. Effect sizes roughly halve under v1.1. The combined p-value moves from 2.4×10^{-7} to 2.9×10^{-6} across five confirmatory chambers (two of which are individually null). Prevalence becomes a floor over a five-month window rather than a corpus rate. **Nothing reverses direction.**

One dependency shifted. F9 was originally reported as "subsumed by F3." With the content-word control removed, F9 stands alone as the justification for v1.1's present-in-both restriction — and it is sufficient on its own: words absent from both periods contribute a constant +0.58 to +0.68 from the smoothing, worth 11% of New Brunswick's raw primary and 31% of the UK's.

2. Findings obviated by removing an analysis

F1 — the mechanism test, and the corpus under it

The attack. §6's strongest anti-p-hacking† defence claimed that a word's "AI preference" — measured from 40 synthetic speeches and a 2019 control — predicts which words later rose in Hansard, at Spearman† +0.548, permutation $p^\dagger < 10^{-4}$. But the 2019 control is a *subset* of the 2018–22 pre-window, so each word's rarity sat inside both the predictor and the outcome with the same sign.

Independently verified, and the mechanism is a variance problem. The predictor was

$$\text{ai_pref}(w) = \log \frac{(\text{syn_n}(w) + 0.5)}{6,838} - \log \frac{(\text{ctl_n}(w) + 0.5)}{703,905}$$

└ rate in AI text ┐ └ rate in 2019 humans ┐

The synthetic corpus was 6,838 words, so **39 of the 89 usable words occurred exactly once** — seventeen distinct values in total, against control counts spanning 0 to 7,759. The variance is 0.758 in the numerator term against 4.390 in the denominator term, so the predictor collapsed into its denominator: it correlates **+0.914** with the control term and **−0.211** with the synthetic term.

quantity	Spearman
published predictor vs the Hansard shift	+0.548
rarity alone: −log(2019 count), no AI data at all	+0.568
AI evidence alone: log(synthetic count)	−0.219
partial correlation† controlling log(pre count)	−0.008
permutation within frequency strata†	p = 0.24

Rarity alone beats the published predictor. The synthetic corpus contributed nothing and slightly hurt.

Response: obviated. The corpus has been removed from the study, so the analysis no longer exists — this is not a defence awaiting correction. What is retained is the design lesson: **an unpaired† comparison of model output against a human corpus inherits that corpus's frequency structure on both sides**. The replacement is paired by construction — an instruct model against its *own base checkpoint* on identical prompts, which cancels baseline frequency exactly (Geng & Trotta; Yakura et al.).

F3 — the content-word control

The attack. Kobak's team annotated all 900 excess words content-versus-style blinded to year; the 462 content words are the field's canonical topic-churn control; no script in the study touched them; and run under v1.0 the primary statistic **cannot tell them apart from the style list** (Canada +0.282 content vs +0.210 style).

Response: obviated, on a ground the review did not consider. Kobak's "content" list is *PubMed's* content — *lachnospiraceae*, *carbapenemase*, *tocilizumab*, *clostridioides*, *retrosigmoid*. Only 105 of 462 appear in legislative text at all. **Whether biomedical jargon moved in a parliament is a fact about parliamentary topics, orthogonal to whether our style measurement is picking up AI here.** The frequency-and-dispersion-matched placebo already controls for in-domain churn, using words that actually occur in the domain, and does it better.

Two things found on the way to that conclusion are worth recording. Under v1.1 the control appears to pass decisively (style − content +0.23 to +0.48), but that is **the pandemic leaving our pre-window**: the steepest declines are *coronavirus* (−4.63), *quarantine* (−4.58), *omicron* (−4.03), *lockdown* (−3.00), and removing just 24% of the list moves Ireland from −0.358 to −0.011. A control that passes because COVID ended is not evidence. And instruct models prefer

Kobak's **content** words too (+0.488 against matched controls), so content was never a clean null — differencing it away removes real signal.

What replaces it is the validation that actually bears on the instrument: does lexicon density predict an independent detector, segment by segment? See §4.

F5a — sensitivity measured on a single 7B generator

Response: obviated. Pangram's catch rate on machine text is far better established publicly than 40 speeches of ours could show — 99.3% on RAID across domains (Dugan et al.), never below 99.8% at Chicago Booth, the only one of four tools to detect reliably in a peer-reviewed VUB evaluation, 1.4% self-reported FNR (Emi & Spero, arXiv: 2402.14873), and training against 19 humanizer tools. Citing that is shorter *and* stronger than reproducing a weaker version with a generator nobody's office uses.

The one claim the corpus was carrying — that the eight-detector null means "little unedited paste" rather than "these detectors don't work here" — now rests on real adjudicated text instead: the same segments those detectors rate unremarkable were confirmed by Pangram as **576 AI-or-Mixed of 643**, and a frontier LLM screen separates its verdicts at AUC† 0.951 while open-weight models manage 0.55–0.72. The null is a property of perplexity-based detection, not of the record.

3. Findings applied

F2 — dispersion. Placebos were matched to instrument words on pre-period frequency only. Corpus linguistics treats **dispersion†** as co-equal (Gries 2008; Egbert & Biber 2019) — a pool of bursty topic words decays faster between windows, depressing the placebo median and inflating every excess. We had written this exact filter for the formality control and never ported it. v1.1 matches on both axes; costs 12–44% of each effect size.

F4 — reference population and stratifier. The NB prevalence frame is 3,998 segments from five sitting months (Oct–Dec 2025, Mar/May 2026) — 10.6% of the corpus, on the steepest part of the ramp. Restated as **"7.5% of New Brunswick segments in sittings from October 2025 to May 2026."** The deciles were of the **Falcon/Binoculars** score, not the Opus screen, and realised AI counts across them were 0,1,1,1,1,0,1,1,1,2 — flat, so the design degenerated to an effective simple random sample. Harmless for the point estimate under equal allocation; it must be said.

F6 — the frequency-weighted secondary. The claim "every cutoff from <1 to <50 per 100k gives $p \leq 0.008$ " was New Brunswick only, unlabelled, and misquoted (NB's <1 cell is 0.015). Full table now printed: **4 of 30 confirmatory cells reach $p < 0.05$.** The equal-weight choice is defended on the merits and is Gray's published design; letting a reader assume the two statistics agree is not defensible.

F7 — formality control. The candidate floor of 200 sat above $\log_2$ bucket 7, but 227 of the instrument's 407 words live in buckets 0–3, so those quotas could never fill: New Brunswick's control was **67 words against 407**, hence systematically *more common* than what it controls. Floor lowered to 20 with neighbour-borrowing; controls now fill 362–407. Gaps barely move (+0.07 to +0.39), so the earlier numbers were under-evidenced rather than wrong.

F8 — the pooled 2024–26 ratio. Chambers weight the post years very differently (Canada is 49% 2024; Ireland 31%), and the markers are non-stationary† inside the window. Per-year series now reported. **Canada's pooled null was a mixture artifact** — it is positive and significant in both 2025 (+0.068) and 2026 (+0.058). See §4 for what else the series revealed.

F9 — smoothing asymmetry. Because the pre-window is 2–3× the post-window, a word absent from *both* periods scores a constant +0.58 to +0.68. The genuinely new part is the placebo side: the pool is built from pre-period counts, so its lowest bucket holds words with count 1, whose expected logFC is about −0.43 — a structural ~1.1 log-unit gap **before any AI exists**. Fixed by the present-in-both restriction, which F9 now justifies on its own.

F11 — citation. Fourth author is **Jan Lause**, not Berens (Berens is the lab, hence the `berenslab` GitHub org). Web-verified: Kobak D, González-Márquez R, Horvát E-Á, Lause J, *Science Advances* 11(27):eadt3813 (2025). Fixed in the methodology and the frozen protocol; **still wrong in the project** `CLAUDE.md` and should be corrected there before it reaches the paper.

F12 — no-interference. **SUTVA**[†] is nowhere in our documents even though §5.2 measures its violation directly. Stating it is free strengthening: contaminated controls bias toward the null, so every reported effect is a floor.

4. Issues the review did not find

These emerged from working through the findings and are, in our judgement, more serious than anything left in the review.

The lexicon is largely a register meter. Per-segment density predicts the Opus detector at $\rho = +0.252$ while frequency- and-dispersion-matched random word sets predict $+0.024$ (0 of 30 beat Kobak) — so the list is genuinely specific. But the association is **the same size before AI existed** ($+0.250$ pre-2023, $n = 20,329$). What the two instruments agree about is a register that predates these tools. That validates Kobak's list *for that register*; it does **not** support segment-level claims of the form "this speech is machine-drafted because it is lexicon-dense."

The detector sees change the lexicon misses. Lexicon density rose 12% pre → post; the Opus score rose 120%. Matching segments on **absolute** density in fine calipers[†] — the two eras agreeing on mean density to two decimal places within each bin — the score gap is **+10.18**, slightly *larger* than the unmatched $+9.85$. Conditioning on density absorbs none of the detector's rise. So the two arms measure different quantities (register shift vs drafting prevalence) and our lexicon effect sizes are conservative.

The instrument decays. Four of six chambers peak and decline across the per-year series, and the US Senate goes negative. Kobak's list was derived from 2024 data, and Geng & Trotta document individual markers collapsing once they become notorious. New Brunswick's lexicon excess more than halves from 2025 to 2026 ($+0.266 \rightarrow +0.105$) while its measured Pangram prevalence **triples** ($4.3\% \rightarrow 13.6\%$). **Trajectory claims must rest on the detector, not the word list.**

F1's replacement has not landed. The paired base-vs-instruct predictor does not survive frequency stratification (pooled $p = 0.098$; neither family replicates alone). The design is sound — its rarity check is $+0.22$ against log-frequency where the dead test scored $+0.92$ — so this is a real null, not the same confound returning. There is still no word-level mechanism evidence.

The likelihood-ratio arm is weak and possibly confounded. Scoring Hansard under instruct-vs-base checkpoints gives small, inconsistent movements (two chambers positive, two null, US Senate significantly negative), against base log-probability changes $3\times$ larger. Instruct models may simply be more robust to unfamiliar post-2023 content, which would produce the same pattern with no register drift.

What did land, and it is worth stating as a contribution: instruct models over-produce Kobak's style words relative to their own base checkpoints at **+0.88, $p < 0.001$** , with 70.8% of style words instruct-preferred against 28.2% of all other words. Kobak inferred LLM causation from a temporal pattern in human text, explicitly avoiding labelled model output; this measures it from the weights, by the route they set out to avoid needing.

5. Glossary

For a reader comfortable with probability and measurement but not with the conventions of empirical social science.

Testing and inference

Null hypothesis. The boring explanation you are trying to rule out. Here it is never "nothing changed" but "this changed no more than comparable vocabulary changed anyway."

p-value. Probability of a result at least this extreme *if the null were true*. Not the probability the null is true. Think of it as a false-alarm rate for a detector thresholded at the observed value.

Permutation / placebo test. Rather than assume a null distribution, build it: re-draw the thing you are measuring many times under conditions where the effect cannot exist, and see where the real value falls. Ours draws 1,000 matched random word sets. It is the empirical-noise-floor approach — measure the noise instead of modelling it.

$p < 0.001$ as a bound. With 1,000 draws the smallest observable p is $1/1000$. " $p < 0.001$ " means zero draws beat the real value — the resolution limit of the simulation, not a measurement.

Stratified permutation. Shuffling only *within* groups sharing a nuisance property (here word frequency), so that structure is preserved and cannot masquerade as signal. A global shuffle destroys it — which is how F1's confound went unseen.

Confound. A third variable driving both things you are relating, producing a correlation that means nothing. In F1 it was the word's own rarity.

Partial correlation. What remains between two variables once a third is removed from both — the residuals correlated against each other. Raw $+0.55$ with a partial of -0.01 means the third variable did all the work.

Spearman (rank) correlation. Pearson correlation computed on *ranks*. Sort the items by each measurement and ask whether the orderings agree. Robust to outliers and to non-linear-but-monotone relationships. Being a within-group quantity, it says nothing about whether the *levels* moved — two instruments can both drift upward while their internal ordering is unchanged.

AUC. Probability that a randomly chosen positive scores above a randomly chosen negative. 0.5 is chance, 1.0 perfect. Threshold-free, so it compares detectors whose score scales differ.

Caliper matching. Comparing groups only within narrow bands of a confounding variable, so the comparison holds it near-constant. Finer than binning by quantile: if a distribution has shifted, one group sits systematically higher *within* each quantile bin, and the comparison is still confounded.

Fisher's method. Combines independent p -values: $X^2 = -2 \cdot \sum \ln(p_i)$ on $2k$ degrees of freedom. Asks whether a pattern is unlikely taken together.

Effect size vs significance. Significance says an effect is unlikely to be noise; effect size says how big. With tens of millions of words everything is significant, so effect size is the only interesting quantity.

Power. Probability of detecting an effect that is really there. A null from a small corpus is weak evidence of absence — which is why the US Senate null on 24M words means something and the earlier 3M-word version did not.

p-hacking / researcher degrees of freedom. Trying many analyses and reporting the one that worked. Defences: fix the analysis before seeing the data, use an instrument someone else built for another purpose, and have no tunable choices in the primary statistic.

Discovery vs confirmatory corpus. The dataset where you *found* an effect cannot also be evidence for it, because you selected on the outcome. New Brunswick is our discovery corpus, quoted separately and as an upper bound.

SUTVA / no-interference. The assumption that one unit's treatment does not affect another's outcome. Violated here in an interesting way — AI writing changes how non-AI text is written — so our controls are partly treated, and every effect is biased toward the null.

Measurement and correction

Sensitivity (Se) = $P(\text{flag} \mid \text{actually AI})$. **Specificity (Sp)** = $P(\text{no flag} \mid \text{actually human})$. Properties of a detector *on your data*, not universal constants.

Rogan–Gladden correction. $\pi = \tau \cdot \text{Se} + (1 - \tau)(1 - \text{Sp})$, so $\hat{\tau} = (\hat{\pi} + \text{Sp} - 1) / (\text{Se} + \text{Sp} - 1)$. With Sp measured at 1, this collapses to $\hat{\tau} = \hat{\pi} / \text{Se}$, and since $\text{Se} \leq 1$, $\hat{\tau} \geq \hat{\pi}$. It contributes no number here — it tells us the *shape* of the uncertainty: one-sided, upward, with $1/\text{Se}$ as the unknown multiplier.

Clopper–Pearson / Wilson intervals. Confidence intervals for a proportion; Clopper–Pearson exact and conservative, Wilson better-behaved near 0 and 1.

Rule of three. Zero events in n trials bounds the rate at about $3/n$. Zero flags in 2,400 pre-LLM segments bounds the false-positive rate near 0.125%.

Two-phase / double sampling. A cheap instrument scores everything and defines strata; the expensive one is spent where informative. The cheap one need not be accurate, only correlated with truth, because stratum weights are known from the full corpus.

Stratifier. The variable used to divide a population before sampling. If uninformative, the design degenerates to simple random sampling — not fatal, but it must be disclosed (F4).

Clustered bootstrap. Resampling whole clusters (here, whole speeches) rather than individual observations, because words within a speech are not independent.

Selection on the outcome. Choosing units by the quantity you then measure in the remainder. Guaranteed to bias, because you removed the upper tail of a noisy distribution. Fixed by **split-half**: rank on one half of the data, measure on the other.

Corpus statistics

Log fold-change. $\log(\text{rate after} \div \text{rate before})$. Symmetric around zero and additive, which is why count ratios are analysed in logs.

Smoothing / pseudocount. Adding a constant (0.5) so zeros do not produce infinities. Not innocent: when the two denominators differ, an absent word gets a systematic non-zero score (F9).

Equal-weight vs pooled. Equal-weight gives each word one vote; pooled sums counts, so common words dominate. They can point opposite ways on the same data — as they do here (F6).

Dispersion. How evenly a word spreads across documents or sittings, as distinct from how often it occurs. 500 occurrences in one debate is frequent but badly dispersed. Ignoring it selects bursty topic words.

Frequency matching. Choosing controls with the same baseline frequency as each instrument word — the analogue of matching impedance before comparing two measurements.

Placebo / donor pool. The set controls are drawn from. Its construction determines what the null represents.

Non-stationary. A process whose statistics change over time. Our markers are non-stationary inside the post-window — some collapsed after 2024, others kept rising — so one pooled ratio over 2.5 years is not a stable quantity.

Paired vs unpaired design. Paired compares two conditions on the *same* units, so unit-level nuisance variation cancels exactly. Base-versus-instruct on identical prompts is paired; model-output-versus-human-corpus is not, and that difference is what killed F1.

Document containment. Kobak's estimand: the fraction of *documents* containing at least one group word, rather than a token rate. It is what licenses a prevalence reading ("at least 13.5%"); a token-rate mean has none. Adding it as a parallel primary is F10, the one outstanding item.